

Lightweight Deep Learning Architecture for Realtime Crop Pest Detection



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Problem & Motivation

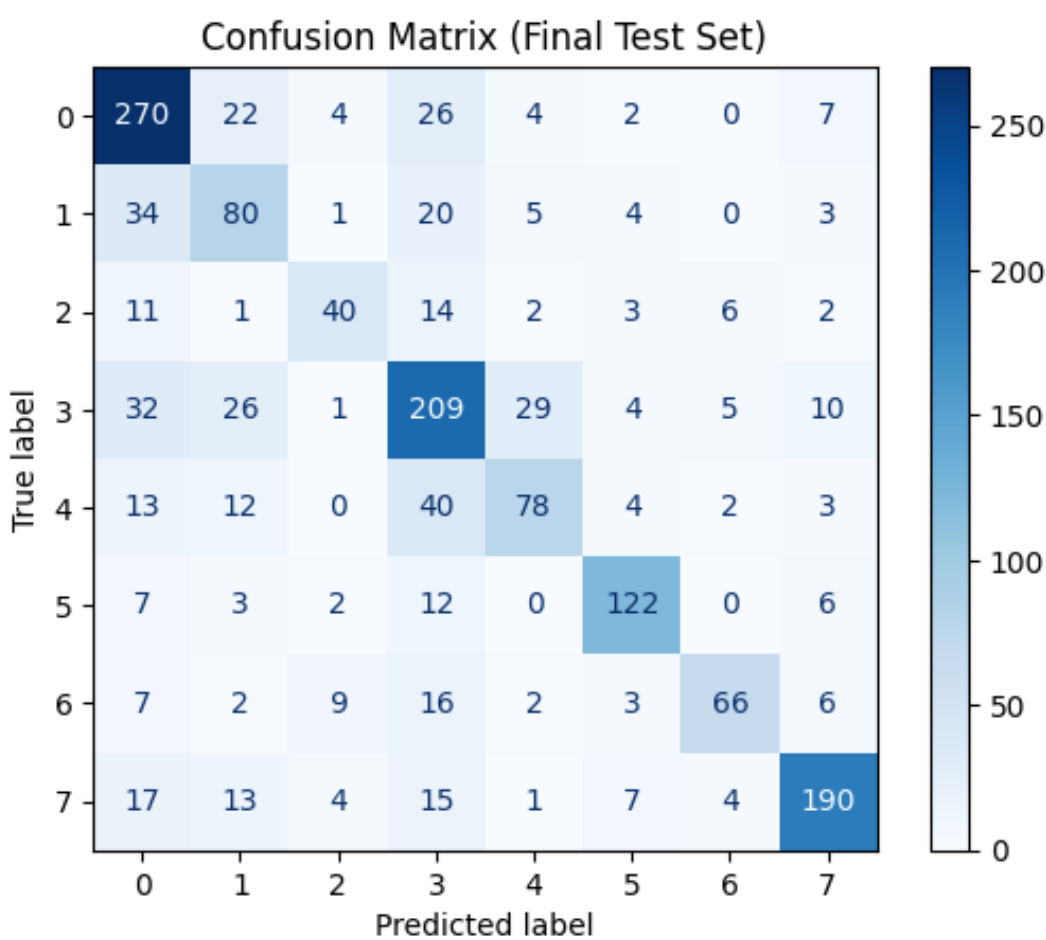
- **Problem Context:** Current state-of-the-art pest detection models lack the architectural optimization needed to achieve high detection accuracy on low-power microcontrollers, drones, and edge devices.
- **Research Gap:** Existing deep learning models lack the accuracy, scalability, and resource efficiency required to run on low-power devices in diverse field conditions.
- **Objective:** Develop a lightweight and efficient deep learning model for real-time crop pest detection that can operate on resource-constrained mobile devices.

Evaluation & Key Results

- Accuracy: **69.0%**
- F1-Score (weighted): **69.0%**

A detailed breakdown of performance by class is shown in the confusion matrix below covering the following classes:

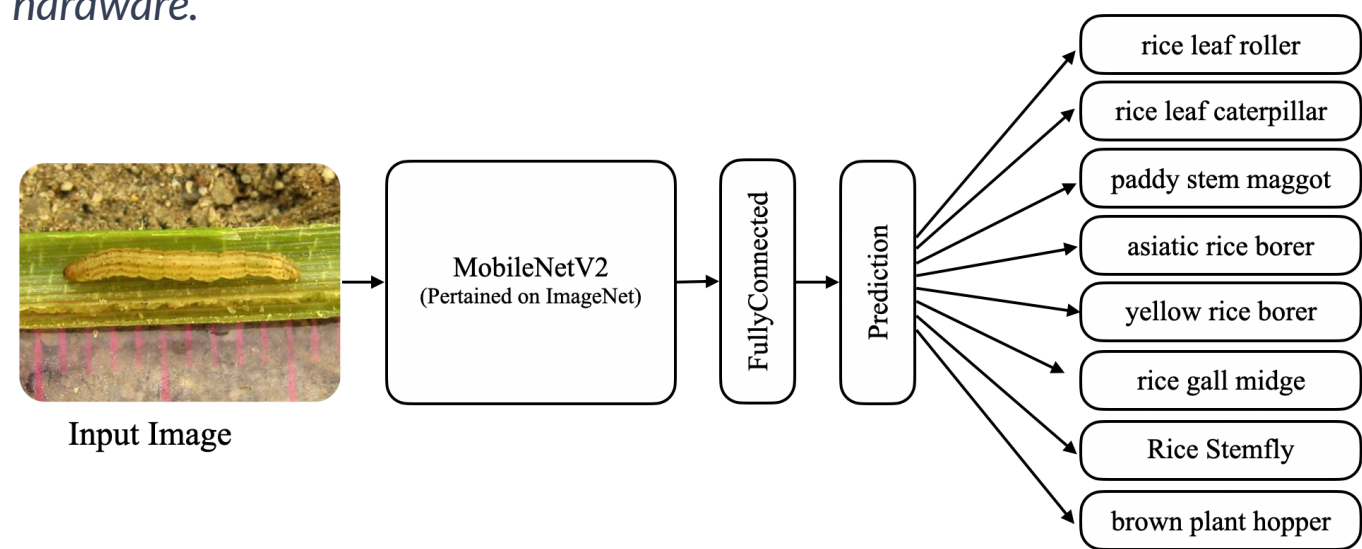
Class Index	Disease Name
0	Rice Leaf Roller
1	Rice Leaf Caterpillar
2	Paddy Stem Maggot
3	Asiatic Rice Borer
4	Yellow Rice Borer
5	Rice Gall Midge
6	Rice Stemfly
7	Brown Plant Hopper



Methodology & Dataset

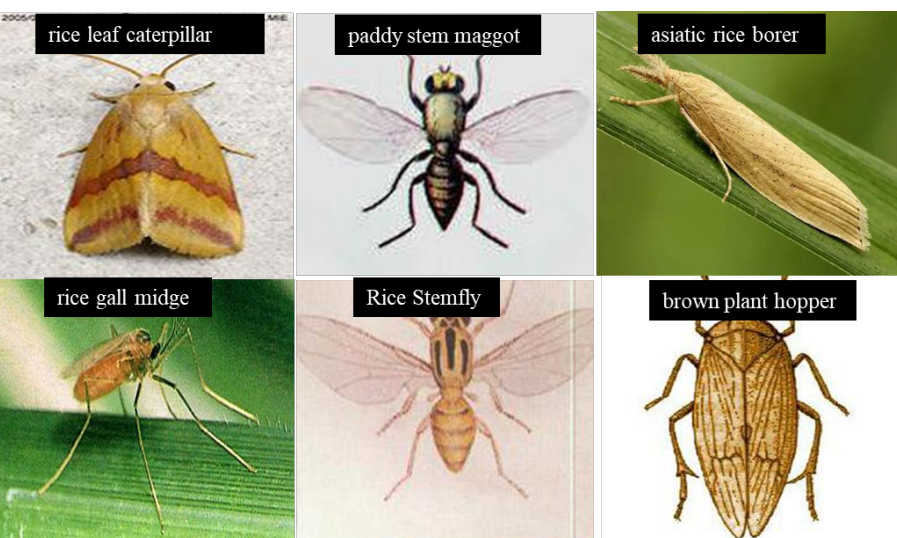
System Architecture

Our LDLA system utilizes an efficient lightweight **MobileNetV2** deep network for real-time crop pest classification. This architecture seamlessly deploys to **edge devices** attached directly to automated waste-sorting hardware.



Dataset

We used the Agricultural Plant Pest Image Dataset (IP102), sourced from Kaggle. The data consists of 75,000 images covering 8 pest categories.



Significance & Conclusion

- **Contribution:** Validated a lightweight deep learning (MobileNetV2) as a practical, computationally efficient solution for real-world crop pest classification in precision agriculture.
- **Impact:** Enables real-time, edge-device pest detection to help farmers identify crop infestations early and minimize yield loss.
- **Future Work:** Future research aims to bridge model development to field deployment by expanding diagnostic capabilities, conducting real-world field testing, and applying advanced optimization techniques to boost accuracy.